

Chapter 9B

Stroke, TIA, Intracranial Hemorrhage, and Vascular Neurology

Medvale Internal Medicine - Neurology for Internal Medicine

Expanded regeneration: this chapter is intentionally content-heavy. It is original Medvale writing using the uploaded stroke/TIA/ICH/SAH source only as a coverage signal, with added diagnostic algorithms, mechanism tables, integrated cases, and Graph-RAG extraction targets. It avoids copying the source figures and replaces them with original tables and decision maps.

1. Why this matters clinically

Stroke is one of the highest-yield neurologic emergencies for internal medicine because management decisions must be made before the diagnosis feels intellectually complete. The clinician must rapidly separate ischemic stroke from hemorrhage, identify candidates for reperfusion, recognize large-vessel occlusion, reverse anticoagulation when bleeding is present, and prevent early complications such as aspiration, cerebral edema, hydrocephalus, recurrent embolization, or hematoma expansion. The goal is not to memorize a list of neurologic syndromes in isolation; the goal is to convert a neurologic deficit into an urgent vascular and anatomic hypothesis.

The practical bedside categories are transient ischemic attack (TIA), acute ischemic stroke, intracerebral hemorrhage (ICH), and subarachnoid hemorrhage (SAH). TIA is a warning syndrome. Ischemic stroke is primarily an arterial occlusion and salvageable-penumbra problem. ICH is a bleeding, hematoma expansion, pressure, and reversal problem. SAH is usually an aneurysm problem with meningeal blood, rebleeding risk, hydrocephalus, vasospasm, and delayed cerebral ischemia.

The uploaded source emphasizes the core Step-style framework: most strokes are ischemic, hemorrhagic strokes are less common but often more immediately dangerous, TIAs may involve carotid or vertebrobasilar territories, and CT head is the rapid first branch point in emergency evaluation. This chapter keeps that coverage while expanding the clinical reasoning around triage, localization, treatment, and secondary prevention.

Core syndrome scripts

Syndrome	Clinical script	Immediate action frame
TIA	Transient focal neurologic dysfunction from ischemia. Symptoms resolve, often within minutes, but early stroke risk is front-loaded in the first hours to days.	Urgent stroke workup: brain imaging, vascular imaging, ECG/telemetry, risk-factor treatment, antiplatelet or anticoagulation plan depending on mechanism.
Acute ischemic stroke	Abrupt focal deficit; CT may be normal early. Deficits may localize to cortical, lacunar, brainstem, cerebellar, or retinal territory.	Noncontrast CT to exclude hemorrhage, CTA for LVO, determine IV thrombolysis and mechanical thrombectomy eligibility, avoid preventable hypoxia, hypoglycemia, fever, and aspiration.
Intracerebral hemorrhage	Focal deficit plus headache, vomiting, depressed consciousness, severe hypertension, or anticoagulant exposure. Acute blood appears hyperdense on CT.	Neurocritical care, BP control, anticoagulant reversal, intracranial pressure assessment, neurosurgical triage.
Subarachnoid hemorrhage	Thunderclap headache, meningismus, photophobia, vomiting, seizure, syncope, or altered mental status. CT can be negative later after onset.	CT/LP/CTA pathway, secure aneurysm, nimodipine, treat hydrocephalus, monitor vasospasm and delayed cerebral ischemia.

2. Neurovascular anatomy and localization

Neurologic localization should be performed before mechanism assignment. A patient with aphasia has a cortical dominant-hemisphere process until proven otherwise. A patient with neglect has nondominant parietal or frontal cortical involvement. A patient with pure motor hemiparesis and no cortical signs has a lacunar syndrome until proven otherwise. A patient with vertigo plus dysarthria, diplopia, dysphagia, ataxia, or crossed sensory/motor findings has posterior circulation disease until proven otherwise.

Localization matters because it changes urgency. A dense cortical syndrome may reflect a proximal MCA occlusion and should trigger CTA/thrombectomy assessment. A basilar syndrome may begin with nonspecific dizziness, nausea, diplopia, or dysarthria but can progress to coma or locked-in syndrome. Cerebellar infarction may appear deceptively mild, then swell and compress the fourth ventricle or brainstem.

Vascular territory localization

Territory	High-yield deficits	Pearl
Dominant MCA	Contralateral face/arm weakness and sensory loss, aphasia, gaze preference, possible homonymous hemianopia.	Aphasia turns a weakness syndrome into a cortical syndrome.
Nondominant MCA	Contralateral weakness and sensory loss, neglect, anosognosia, constructional difficulty.	The patient may deny the deficit, creating a false sense of reassurance.
ACA	Contralateral leg greater than arm weakness, abulia, grasp reflex, urinary incontinence.	Think medial frontal/paracentral lobule.
PCA	Contralateral homonymous hemianopia, visual agnosia, memory deficits, alexia without agraphia in selected dominant lesions.	Isolated visual complaint in a vascular-risk patient can be stroke.
Vertebrobasilar	Diplopia, dysarthria, dysphagia, vertigo, ataxia, crossed cranial nerve/long tract signs, altered consciousness.	Dizziness alone is common; dizziness plus brainstem signs is dangerous.
Cerebellum	Severe gait ataxia, dysmetria, nystagmus, vomiting, headache.	Inability to sit or walk independently is a major red flag.
Internal capsule	Pure motor hemiparesis; sometimes dysarthria-clumsy hand.	Dense motor deficit without aphasia, neglect, or visual field loss.
Thalamus	Pure sensory loss, central pain syndrome, altered arousal if bilateral.	Small lesion can cause prominent sensory syndrome.

3. Mechanism-first thinking: why the vessel closed

The mechanism determines prevention. Large-artery plaque, atrial fibrillation, lacunar lipohyalinosis, dissection, paradoxical embolism, endocarditis, vasculitis, malignancy-associated thrombosis, and stimulant-induced vasospasm can all cause ischemic stroke, but they require different prevention strategies. A patient with atrial fibrillation needs anticoagulation when safe; a patient with symptomatic severe carotid stenosis may need carotid revascularization; a patient with lacunar disease needs aggressive hypertension and diabetes control; a patient with dissection may need antithrombotic therapy and vascular follow-up.

Stroke mechanisms

Mechanism	Source/pathology	Suggestive clues	Prevention logic
Large-artery atherosclerosis	Plaque in carotid, vertebral, basilar, or intracranial artery with thrombus or artery-to-artery embolus.	Carotid bruit, amaurosis fugax, cortical deficit, recurrent events in same territory.	Antiplatelet, statin, risk factor control; selected symptomatic carotid stenosis may need endarterectomy/stenting.
Cardioembolism	Embolus from atrial fibrillation, LV thrombus, valvular disease, endocarditis, cardiomyopathy, PFO with venous source.	Sudden maximal deficit, multiple vascular territories, cortical involvement.	Anticoagulation when safe if AF/strong cardioembolic indication; treat structural source.
Lacunar disease	Small penetrating artery disease from lipohyalinosis, usually HTN/DM related.	Pure motor, pure sensory, ataxic hemiparesis, dysarthria-clumsy hand.	BP and diabetes control, statin, antiplatelet unless contraindicated.
Hypoperfusion	Low-flow state, severe stenosis, shock, cardiac arrest, excessive BP lowering.	Watershed pattern; bilateral border-zone deficits.	Restore perfusion; avoid overcorrection of BP in selected patients.
Dissection	Intimal tear in carotid or vertebral artery.	Neck pain/headache, trauma/manipulation, Horner syndrome, posterior symptoms.	Antithrombotic therapy and vascular follow-up; selected interventions.
Unusual/young stroke	Vasculitis, APS, malignancy, sickle cell, drug-induced vasospasm, thrombophilia.	Young age, systemic symptoms, miscarriages, rash, cancer, cocaine/amphetamine use.	Targeted testing and mechanism-specific treatment.

4. TIA: transient symptoms, non-transient risk

A TIA should not be dismissed because the exam normalizes. A TIA means that a vascular mechanism became clinically active and may repeat. The risk is highest soon after the event, especially with motor weakness, speech disturbance, longer symptom duration, diabetes, hypertension, atrial fibrillation, symptomatic carotid disease, or crescendo episodes.

The classic time-based definition uses resolution within 24 hours, but modern practice increasingly emphasizes tissue-based classification because MRI may show infarction despite symptom resolution. Clinically, the emergency question is unchanged: treat TIA as an urgent stroke-prevention opportunity.

ABCD2 risk score components

Component	Points	Clinical meaning
Age ≥ 60 years	1	Older age increases baseline vascular risk.
Blood pressure $\geq 140/90$ at presentation	1	Marker of active vascular stress.
Unilateral weakness	2	Highest-risk clinical feature in the score.
Speech impairment without weakness	1	Cortical ischemic symptom.
Duration ≥ 60 minutes	2	Longer ischemia increases risk.
Duration 10-59 minutes	1	Intermediate risk.
Diabetes mellitus	1	Systemic vascular risk amplifier.

ABCD2 is not a rule-out tool. A patient with crescendo TIAs, atrial fibrillation, monocular blindness, posterior circulation symptoms, or symptomatic carotid stenosis can be high risk even with a modest score.

5. Initial emergency assessment

Emergency stroke assessment is parallel processing. While one clinician obtains the last-known-well time and neurologic exam, another should check glucose, establish IV access, review anticoagulants, and move the patient toward noncontrast CT. The first imaging question is: is there blood? The second is: is there a treatable large-vessel occlusion? The third is: how much tissue is already infarcted and how much may be salvageable?

- Obtain exact last-known-well time, not the time symptoms were discovered.
- Check bedside glucose immediately because hypoglycemia can mimic stroke.
- Measure NIH Stroke Scale to quantify severity and track change.
- Ask about anticoagulants, recent surgery, bleeding, trauma, seizure at onset, and prior intracranial hemorrhage.
- Noncontrast CT head is the first urgent imaging test in most emergency pathways.
- CTA head/neck is needed when large-vessel occlusion, dissection, aneurysm, or vascular stenosis is suspected.
- Do not delay time-sensitive reperfusion decisions for low-yield etiologic testing.

Acute imaging and testing

Test	Best use	Key limitation
Noncontrast CT head	Rapid exclusion of hemorrhage, mass, hydrocephalus, large established infarct, subdural hematoma.	Can be normal early in ischemic stroke; posterior fossa lesions may be subtle.
CTA head/neck	Large-vessel occlusion, carotid/vertebral stenosis, dissection, aneurysm, vascular malformation.	Requires contrast and interpretation; does not alone prove tissue viability.
CT/MR perfusion	Selected extended-window thrombectomy triage; estimates core and penumbra.	Thresholds can be artifact-prone; not a substitute for clinical judgment.
MRI DWI/ADC	Highly sensitive infarct detection, lacunar lesions, posterior fossa lesions, stroke mimics.	Availability, time, implants, agitation; early posterior circulation false negatives can occur.
ECG/telemetry	Atrial fibrillation, acute MI, rhythm clues for embolism.	Single ECG does not exclude paroxysmal AF.
Echocardiography	LV thrombus, valvular disease, PFO, endocarditis, structural source.	Yield depends on patient selection; TEE may be needed for selected sources.

6. Original acute stroke triage map

Acute triage map

Branch	Action
Suspected acute stroke	Parallel tasks: glucose, NIHSS, last-known-well, anticoagulant history, IV access, noncontrast CT.
CT shows hemorrhage	ICH/SAH pathway: ICU, BP control, anticoagulant reversal, CTA/vascular imaging when indicated, neurosurgical/neurocritical care.
CT shows no hemorrhage	Ischemic pathway: determine IV thrombolysis eligibility and CTA for large-vessel occlusion.
Large-vessel occlusion present	Activate thrombectomy pathway if imaging/time/clinical criteria fit; transfer if needed.
No LVO or not eligible	Medical ischemic stroke care: antiplatelet timing, statin, swallow screen, VTE prevention, rehab, etiology workup.
Symptoms resolved	TIA/minor stroke pathway: urgent imaging, short-term recurrence prevention, rhythm and carotid evaluation.

7. Acute ischemic stroke treatment

Acute ischemic stroke treatment has several layers: supportive stabilization, reperfusion for eligible patients, prevention of early worsening, and secondary prevention. The 2019 AHA/ASA acute ischemic stroke guideline supports rapid evaluation for IV thrombolysis in eligible patients and endovascular thrombectomy for selected patients with large-vessel occlusion. Eligibility details are institution-specific and should be protocol-driven, but the reasoning framework is stable: exclude hemorrhage, define time, identify disabling deficit, identify LVO, and avoid preventable physiologic injury.

Treatment layers in acute ischemic stroke

Layer	Reasoning	Examples
Stabilization	Neurons fail faster with hypoxia, hypoglycemia, severe hyperglycemia, fever, hypotension, or aspiration.	Airway/oxygenation, glucose correction, temperature control, swallow screening.
Reperfusion	Open the occluded artery when benefit outweighs bleeding risk.	IV thrombolysis in selected patients; mechanical thrombectomy in selected LVO.
Antithrombotic timing	Prevent early recurrence while avoiding hemorrhagic transformation.	Aspirin after hemorrhage excluded; delay after thrombolysis per protocol.
Complication prevention	Stroke units reduce death and disability through standardized prevention.	DVT prophylaxis, mobilization, dysphagia care, pressure injury prevention.
Mechanism workup	Prevention depends on cause.	Telemetry, vascular imaging, echo, lipids/A1c, selective hypercoagulable/vasculitis testing.

High-yield treatment trap: anticoagulation is not an acute salvage treatment for most ischemic strokes. It is a secondary prevention tool for selected mechanisms such as atrial fibrillation, started when hemorrhagic transformation risk is acceptable.

8. Blood pressure, glucose, fever, and oxygen

Physiologic management is not glamorous but is central. In ischemic stroke, blood pressure may support collateral flow, so aggressive lowering can worsen penumbral ischemia unless there is a competing emergency or the patient needs thrombolysis. If thrombolysis is planned, blood pressure must be below protocol thresholds before and after treatment. Fever increases metabolic demand and worsens outcomes; hypoxia reduces oxygen delivery; severe hyperglycemia is associated with worse ischemic injury, while hypoglycemia is a mimic and an injury amplifier.

Physiology targets: concept-level anchors

Variable	Concept	Clinical implication
Blood pressure	May support collateral perfusion in ischemic stroke; can worsen bleeding in ICH.	Permissive hypertension may be appropriate in ischemia; smooth reduction is important in hemorrhage.
Glucose	Hypoglycemia mimics stroke; severe hyperglycemia worsens outcome.	Check immediately; correct extremes.

Variable	Concept	Clinical implication
Temperature	Fever increases metabolic demand and worsens brain injury.	Treat fever and search for infection or aspiration.
Oxygenation	Hypoxia accelerates neuronal injury.	Give oxygen for hypoxemia, not as a substitute for reperfusion.
Swallow function	Dysphagia causes aspiration pneumonia.	Screen before oral intake.

9. Early complications of ischemic stroke

Stroke often worsens after the first scan. Large infarcts can swell over 1 to 5 days, with malignant MCA infarction causing midline shift and herniation. Cerebellar infarcts can cause posterior fossa swelling, obstructive hydrocephalus, and brainstem compression. Hemorrhagic transformation is more likely with large infarcts, reperfusion, uncontrolled hypertension, and anticoagulation. Medical complications include aspiration pneumonia, DVT/PE, urinary retention, pressure injury, delirium, depression, and deconditioning.

- Malignant MCA infarction: declining consciousness after large MCA infarct; consider decompressive hemicraniectomy in selected patients.
- Cerebellar infarct: vomiting, severe ataxia, headache; monitor for hydrocephalus and brainstem compression.
- Hemorrhagic transformation: neurologic worsening after ischemic stroke; repeat imaging urgently.
- Aspiration: dysphagia screening before oral intake is a prevention intervention.
- Seizure: treat clinical seizures; routine prophylaxis is not universal.

10. Secondary prevention: match treatment to mechanism

Secondary prevention begins once the acute trajectory is safe. Mechanism matters more than a generic label. A noncardioembolic stroke usually requires antiplatelet therapy, high-intensity statin, blood pressure management, diabetes care, smoking cessation, and lifestyle modification. A cardioembolic stroke from atrial fibrillation usually requires anticoagulation when safe. Symptomatic severe carotid stenosis may benefit from revascularization. Cryptogenic stroke requires deliberate rhythm and structural evaluation rather than reflexive rare testing.

Secondary prevention by mechanism

Mechanism	Prevention strategy	Common error
Large artery atherosclerosis	Antiplatelet, statin, BP/diabetes/smoking control; selected carotid revascularization.	Treating symptomatic carotid stenosis as a medication-only problem when revascularization is indicated.
Atrial fibrillation	Anticoagulation when safe; assess timing based on infarct size and bleeding risk.	Starting too early after large infarct or forgetting long-term rhythm monitoring.
Lacunar stroke	Aggressive BP/DM/lipid control and antiplatelet therapy unless contraindicated.	Over-testing for rare causes without clues.
Dissection	Antithrombotic strategy and vascular follow-up.	Missing neck pain/headache or recent manipulation.
Cryptogenic/ESUS	Extended rhythm monitoring, echo, vascular imaging, selected labs.	Assuming cryptogenic means no cause exists.

11. Intracerebral hemorrhage

Intracerebral hemorrhage is bleeding into the brain parenchyma. Hypertension classically causes deep hemorrhage in the basal ganglia, thalamus, pons, or cerebellum. Lobar hemorrhage in older adults suggests cerebral amyloid angiopathy, but anticoagulation, tumor, AVM, cavernous malformation, vasculitis, and hemorrhagic transformation of infarct are also important. The danger is not only the initial bleed; hematoma expansion, edema, intraventricular extension, hydrocephalus, seizures, and increased ICP can cause rapid deterioration.

ICH location patterns

Location	Common association	Clinical clue
Putamen/basal ganglia	Chronic hypertension.	Contralateral weakness/sensory loss; common deep hypertensive site.
Thalamus	Hypertension.	Sensory deficits, eye movement abnormalities, altered arousal if large.
Pons	Hypertension.	Coma, quadriparesis, pinpoint pupils; very high risk.
Cerebellum	Hypertension, vascular lesion.	Vomiting, ataxia, headache; hydrocephalus/brainstem compression.
Lobar cortex/subcortex	Amyloid angiopathy, anticoagulation, tumor, AVM.	Seizures and cortical deficits; recurrent lobar hemorrhage risk.

ICH management framework

- Admit significant ICH to ICU/neurocritical setting.
- Reverse anticoagulation urgently when present: warfarin generally requires PCC plus vitamin K; DOAC reversal depends on agent and local protocol.
- Lower severe blood pressure smoothly; avoid sudden hypotension that can reduce cerebral perfusion.
- Treat increased ICP with head elevation, analgesia/sedation, osmotic therapy, ventricular drainage, and neurosurgical strategies when indicated.
- Evaluate cerebellar hemorrhage urgently because surgery can be lifesaving with deterioration, brainstem compression, or hydrocephalus.
- Search for secondary causes when location, age, imaging, or history is atypical.

ICH pearl: 'blood pressure control' does not mean reflexive aggressive hypotension. The concept is controlled, smooth reduction while preserving cerebral perfusion.

12. Subarachnoid hemorrhage

Aneurysmal SAH usually presents with a thunderclap headache that reaches maximal intensity rapidly. The headache may be accompanied by neck stiffness, photophobia, vomiting, syncope, seizure, or altered mental status. A warning leak can produce a sentinel headache days before rupture. The most important immediate problem is diagnosing the bleed and securing the aneurysm before rebleeding.

Noncontrast CT is the first test. If CT is unrevealing but the story remains highly concerning, diagnostic pathways include lumbar puncture for xanthochromia/RBC pattern and/or vascular imaging depending on time from onset, local practice, and patient-specific factors. Papilledema, mass effect, or focal mass lesion should be considered before LP.

SAH complications and management anchors

Complication	Mechanism	Management anchor
Rebleeding	Unsecured aneurysm reruptures.	Early aneurysm securing by coiling or clipping.
Delayed cerebral ischemia/vasospasm	Arterial narrowing and downstream ischemia days after SAH.	Nimodipine; euvolemia; treat symptomatic DCI in monitored setting.
Hydrocephalus	Blood obstructs CSF flow.	External ventricular drain when indicated.
Hyponatremia	SIADH or cerebral salt wasting.	Monitor sodium and volume status; avoid hypovolemia.
Seizures	Blood irritates cortex.	Treat seizures; prophylaxis individualized.

13. Stroke mimics and traps

Common stroke mimics

Mimic	Why it resembles stroke	Clue or safeguard
Hypoglycemia	Can cause focal deficits and altered mental status.	Check glucose before committing to diagnosis.

Mimic	Why it resembles stroke	Clue or safeguard
Seizure with Todd paralysis	Postictal focal weakness can persist.	Witnessed seizure, tongue bite, postictal state; still image if uncertain.
Migraine aura	Visual/sensory/language symptoms.	Positive spreading symptoms; first/worst or vascular-risk cases still require evaluation.
Subdural hematoma	Fluctuating focal deficits, confusion.	Falls, anticoagulation, older age; CT detects.
Functional neurologic disorder	Weakness/sensory symptoms can be dramatic.	Inconsistency helps, but do not diagnose before dangerous mimics excluded.
Vestibular neuritis	Dizziness, nausea, nystagmus.	Can be confused with posterior circulation stroke; inability to walk or brainstem signs are red flags.

High-yield posterior circulation trap: vertigo plus diplopia, dysarthria, dysphagia, truncal ataxia, direction-changing nystagmus, new severe headache, or inability to walk is not routine benign vertigo.

14. Young stroke and atypical stroke workup

Young stroke requires a wider but still clue-driven differential. The uploaded source highlights additional testing in younger patients, including thrombophilia and vasculitis considerations. The goal is not to send every rare test on every patient; it is to identify mechanisms that are more common when atherosclerosis and atrial fibrillation are less likely.

Young/atypical stroke clues

Clue	Consider	Testing direction
Neck pain, headache, trauma/manipulation	Carotid or vertebral dissection.	CTA/MRA neck.
Miscarriages, thrombosis, livedo, autoimmune symptoms	Antiphospholipid syndrome or autoimmune disease.	APS labs, ANA/ESR/CRP when appropriate.
Fever, murmur, IV drug use	Infective endocarditis with septic emboli.	Blood cultures, echo.
Cancer, weight loss, multifocal infarcts	Malignancy-associated hypercoagulability.	Cancer evaluation guided by history/exam.
Cocaine/amphetamine use	Vasospasm, hypertension, dissection, ICH.	Toxicology plus vascular imaging as indicated.
Sickle cell disease	Large-vessel vasculopathy or anemia-related ischemia.	Hematology/stroke protocol evaluation.

15. Integrated cases

Case 1: acute aphasia

A 76-year-old with atrial fibrillation develops sudden right face/arm weakness and aphasia 70 minutes ago. CT shows no hemorrhage.

Reasoning: This is a dominant left MCA syndrome, likely embolic. The immediate branch is IV thrombolysis eligibility plus CTA for LVO/thrombectomy pathway. Long-term prevention will likely involve anticoagulation when safe.

Case 2: transient monocular blindness

A 68-year-old has 8 minutes of curtain-like vision loss in the right eye and a right carotid bruit.

Reasoning: Amaurosis fugax suggests ipsilateral carotid embolization. This is a high-risk TIA pattern requiring urgent carotid imaging, antiplatelet/statin/risk control, and revascularization consideration if symptomatic severe stenosis is confirmed.

Case 3: pure motor hemiparesis

A 58-year-old with long-standing hypertension has pure left motor weakness without aphasia, neglect, or visual field loss.

Reasoning: This localizes to internal capsule or pons and suggests lacunar small-vessel disease. Prevention focuses on BP, diabetes, lipids, and antiplatelet therapy rather than anticoagulation unless another indication exists.

Case 4: cerebellar danger

A 64-year-old has severe vertigo, vomiting, inability to walk, occipital headache, and nystagmus.

Reasoning: Posterior circulation/cerebellar stroke must be excluded. Cerebellar infarcts can swell and obstruct the fourth ventricle; CT can be insensitive early, so MRI/CTA and close monitoring are often needed.

Case 5: deep hemorrhage

A 61-year-old with severe hypertension develops vomiting, depressed consciousness, and a right putaminal hyperdensity on CT.

Reasoning: This is likely hypertensive ICH. Management prioritizes ICU care, smooth BP control, reversal of coagulopathy if present, ICP vigilance, and repeat imaging/neurologic monitoring.

Case 6: thunderclap headache

A 47-year-old has sudden worst headache of life with neck stiffness. CT performed 18 hours later is negative.

Reasoning: SAH is still possible. Diagnostic pathway should continue with LP for xanthochromia/RBC interpretation and/or vascular imaging depending on local protocol.

16. High-yield exam clues

- Most strokes are ischemic; hemorrhagic strokes are less common but more likely to require reversal, BP, ICU, and neurosurgical triage.
- MCA stroke causes contralateral face/arm weakness; dominant lesions cause aphasia; nondominant lesions cause neglect.
- Lacunar syndromes are focal and stereotyped: pure motor, pure sensory, ataxic hemiparesis, dysarthria-clumsy hand.
- Vertebrobasilar stroke causes cranial nerve symptoms, ataxia, dysarthria, diplopia, dysphagia, or crossed findings.
- TIA involving monocular blindness suggests carotid disease until proven otherwise.
- ICH appears hyperdense on acute CT; ischemic stroke can have a normal early CT.
- SAH classically causes thunderclap headache and xanthochromia when CSF blood has aged enough.
- Cerebellar hemorrhage or infarct can kill through brainstem compression or hydrocephalus despite a small-appearing focal syndrome.
- Do not give antithrombotics until hemorrhage is excluded.
- Do not rapidly lower BP in ischemic stroke unless a protocol or competing emergency requires it.

17. Graph-RAG extraction map

Concept nodes and relationships

Node	Class	Edges to encode
Acute ischemic stroke	Disease	caused_by arterial occlusion; diagnosed_by CT/CTA/MRI; treated_by thrombolysis/thrombectomy/antiplatelet/statin; complicated_by edema/hemorrhagic transformation.
TIA	Warning syndrome	predicts ischemic stroke; evaluated_by brain imaging/vascular imaging/ECG; treated_by secondary prevention.
Large vessel occlusion	Vascular lesion	causes cortical deficits; diagnosed_by CTA/MRA; treated_by mechanical thrombectomy in selected patients.
Atrial fibrillation	Risk factor	causes cardioembolic stroke; treated_by anticoagulation when safe.
Carotid stenosis	Vascular lesion	causes amaurosis fugax/TIA/stroke; evaluated_by ultrasound/CTA/MRA; treated_by endarterectomy/stenting in selected symptomatic high-grade stenosis.
Lacunar infarct	Subtype	associated_with HTN/DM; localizes_to internal capsule/thalamus/pons; presents_as pure motor/pure sensory/ataxic hemiparesis.
Intracerebral hemorrhage	Hemorrhagic stroke	caused_by HTN/amyloid/anticoagulation/AVM/tumor; treated_by BP control/reversal/ICP management/neurosurgery.
Subarachnoid hemorrhage	Hemorrhagic stroke	presents_as thunderclap headache; diagnosed_by CT/LP/CTA; complicated_by rebleeding/vasospasm/hydrocephalus; treated_by coiling/clipping/nimodipine.
Nimodipine	Therapy	reduces delayed cerebral ischemia after aneurysmal SAH.
CTA head/neck	Diagnostic test	detects LVO/dissection/aneurysm/stenosis; guides thrombectomy and SAH workup.

18. Practice questions

Q1. A patient has aphasia and right arm weakness. Which vascular territory is most likely?

Answer: Dominant MCA.

Q2. A patient with resolved unilateral weakness and diabetes had symptoms for 70 minutes. What is the clinical importance?

Answer: High-risk TIA; urgent stroke workup and prevention.

Q3. Why is noncontrast CT performed first in acute stroke?

Answer: To identify hemorrhage and other contraindications before antithrombotic or reperfusion decisions.

Q4. Which syndrome suggests lacunar stroke?

Answer: Pure motor hemiparesis without cortical signs.

Q5. Which hemorrhage is associated with thunderclap headache and xanthochromia?

Answer: Subarachnoid hemorrhage.

Q6. What medication is used after aneurysmal SAH to reduce delayed cerebral ischemia?

Answer: Nimodipine.

Q7. What is the most important next step before LP in a patient with papilledema and possible SAH?

Answer: Repeat/obtain neuroimaging first because LP may precipitate herniation if mass effect is present.

Q8. A cerebellar stroke patient becomes somnolent. What complication is feared?

Answer: Posterior fossa swelling with brainstem compression or obstructive hydrocephalus.

19. References and guideline cross-check anchors

This chapter was generated as original educational content using the uploaded neurovascular source as a coverage guide. It was cross-checked against AHA/ASA guidance for acute ischemic stroke, spontaneous intracerebral hemorrhage, and aneurysmal subarachnoid hemorrhage. Local institutional protocols should determine exact thrombolysis eligibility, thrombectomy imaging thresholds, blood pressure targets, anticoagulant reversal pathways, and neurosurgical escalation.

Medvale Graph-RAG note: this chapter is designed to be chunked into nodes for vascular territories, mechanisms, diagnostics, treatments, complications, and prevention relationships rather than stored only as a linear narrative.